

Wage Transparency and Within-Rank Salary Compression: Evidence from UC Faculty Compensation, 2012–2024

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Abstract

We examine whether the 2022 increase in the salience of wage information in the University of California (UC) system corresponds with measurable changes in within-rank faculty salary dispersion. Using a 13-year repeated cross-section of approximately 229,000 salary records from 2012 to 2024—covering five faculty ranks across ten UC campuses and the UC Office of the President—we construct annual coefficients of variation (CV) for each rank and estimate log-linear OLS regressions with rank and campus fixed effects and a linear time trend. The baseline regression yields a *post2022* coefficient of -0.063 ($p < 0.001$), indicating wages were approximately 6.3% below the secular trend from 2022 onward. The CV declined for Full (-2.1 pp), Associate (-4.6 pp), Research (-2.1 pp), and Clinical (-0.9 pp) professors post-2022, while the Assistant Professor CV rose by roughly 1.7 percentage points, consistent with external labor-market pressures. A difference-in-differences interaction model finds no significant narrowing of the wage gap between Full and Assistant professors, though Associate and Clinical professors fared relatively better in the post-period. We interpret these patterns as consistent with transparency-related internal equity adjustments concentrated *within* rather than *across* faculty ranks.

1 Introduction

The University of California (UC) system is one of the largest public employers in the United States, employing over 265,000 faculty and staff across ten campuses and the UC Office of the President (University of California, 2025). Compensation within the UC system has long been subject to a high degree of transparency. Following a 2008 court decision, the *Sacramento Bee* created a searchable online database allowing the public to view the salary of any California state employee, including UC faculty (Card et al., 2012, p. 2982). Since that time, multiple websites have made UC salary data easily accessible, allowing employees to compare their compensation with that of their peers.

This institutional setting provides a useful environment to study the relationship between wage transparency and wage compression. Economic theory suggests that when workers are able to observe the compensation of their peers, disparities within comparable groups may generate dissatisfaction and reduce job satisfaction among lower-paid employees (Card et al., 2012, p. 2981). Employers may therefore face incentives to narrow wage gaps within job categories in order to retain workers and maintain workplace stability (Cullen & Pakzad-Hurson, 2022). In this paper we examine whether sustained transparency, combined with recent policy changes, corresponds with measurable changes in wage dispersion within faculty ranks in the UC system.

Our analysis focuses on five faculty categories: Assistant, Associate, Full, Clinical, and Research professors. Using thirteen years of compensation data from 2012 through 2024, covering approximately 229,000 professor-year observations, we examine within-rank wage dispersion using several measures including the coefficient of variation (CV), the interquartile range, and the standard deviation.

A key focus of the paper is the period beginning in 2022, which represents a major shift in the transparency and compensation environment facing UC employees. California’s Pay Transparency Act took effect in January 2023, expanding wage disclosure requirements for employers across the state. Around the same time, the UC system experienced its largest academic labor action in history. In late 2022, tens of thousands of academic workers went on strike over wages and working conditions, resulting in significant contract settlements for lower-paid employees. Together, these developments substantially increased the salience of wage information and public discussion of pay disparities within the UC system.

While salary information had technically been public for years, the events surrounding 2022 made wage comparisons far more visible and politically salient. This paper therefore examines whether this period corresponds with measurable changes in within-rank salary dispersion among UC faculty.

The remainder of the paper proceeds as follows. Section 2 outlines the theoretical framework linking wage transparency to compensation compression. Section 3 describes the data sources. Section 4 presents the empirical strategy used to measure wage dispersion. Sections 5 and 6 present the results and discussion, and Section 7 concludes.

2 Theoretical Framework

This framework motivates our empirical investigation of the post-2022 policy environment in the University of California faculty labor market. The 2022 transparency legislation and labor-action event substantially increased the salience of wage comparisons, providing context for the patterns we document. Economic research shows that when workers can observe the compensation of their peers, relative pay differences become an important factor shaping worker satisfaction and effort.

As demonstrated in Bandiera et al. (2005) and Card et al. (2012), workers who discover that they are paid less than comparable peers experience a decline in utility due to perceived unfairness in compensation. Conceptually, when compensation information is opaque the salience of wage comparisons is minimal. However, when compensation becomes highly visible, disparities between comparable workers become more likely to affect worker behavior.

Following Fehr & Schmidt (1999), we represent worker utility as

$$U_i = W_i - C(e_i) - \lambda \max(0, W_{\text{peer}} - W_i), \quad (1)$$

where W_i represents wages, $C(e_i)$ is the convex cost of effort, and the final term captures the disutility experienced when a worker observes a higher-paid peer. The parameter λ represents the salience of wage comparisons. When compensation is opaque, $\lambda \approx 0$. When transparency increases, λ rises, and workers become more sensitive to pay disparities within their reference group.

If wage transparency increases the salience of pay disparities, employers face pressure to reduce wage gaps within comparable job categories in order to prevent dissatisfaction and maintain effort. Cullen & Pakzad-Hurson (2022) show that transparency can fundamentally alter equilibrium wage-setting by making individualized pay differences highly visible and therefore costly for employers to sustain. In response, firms may reduce within-group wage dispersion in order to limit internal dissatisfaction.

Applied to the UC faculty labor market, this framework predicts that transparency pressures should operate primarily within comparable ranks rather than across structurally different positions. An Assistant Professor and a Full Professor occupy distinct positions in

the academic hierarchy, reflecting differences in tenure, experience, and human capital. As a result, transparency-driven compression is expected to occur primarily among direct peers within the same rank.

3 Data

To examine the evolution of faculty compensation, we constructed a 13-year repeated cross-section dataset covering the period from 2012 through 2024. The salary data come from two public administrative sources: the UC Annual Wage Database for 2012 and the California State Controller’s Government Compensation in California database for 2013–2024. After deduplication and filtering to observations with positive total wages, the merged dataset contains 229,386 professor-year observations.

Each annual dataset was standardized to five core variables used in our analysis: year, employer name, position title, regular pay, and total wages. To isolate the relevant academic labor market, we restricted the dataset to records whose position title matched a professor pattern, capturing titles containing “Prof” while excluding administrative or professional staff roles.

Because position titles in the raw administrative data appear as free-text strings, we constructed a rule-based mapping procedure to classify titles into five faculty ranks: Assistant, Associate, Clinical, Research, and Full professors. Titles containing “assistant” or “asst” were classified as Assistant Professors, those containing “associate” or “assoc” as Associate Professors, titles containing “clinical” or “clin” as Clinical Professors, and titles containing “research” or “res” as Research Professors. Remaining titles containing “professor” were classified as Full Professors.

Segmenting the data by faculty rank is essential for our empirical analysis. Differences between ranks largely reflect structural differences in position and experience rather than true inequality within comparable roles. By grouping faculty into comparable ranks, we are able to isolate the dispersion of wages among peers occupying similar academic positions.

Prior to finalizing the dataset, we conducted several quality checks. Duplicate records were removed, observations with negative pay values were excluded, and missing values were filtered out to ensure the integrity of the analysis.

4 Empirical Strategy

To examine whether the post-2022 policy environment corresponds with changes in faculty compensation, we estimate a log-linear Ordinary Least Squares (OLS) regression model.

Using a logarithmic transformation of total wages allows the coefficients to be interpreted as approximate percentage changes.

Our primary specification is

$$\log(\text{TotalWages}_i) = \beta_0 + \beta_1 \cdot \text{post2022}_i + \beta_2 \cdot \text{Rank}_i + \beta_3 \cdot \text{Campus}_i + \beta_4 \cdot \text{Year}_i + \varepsilon_i. \quad (2)$$

The indicator variable `post2022` equals one for 2022 and later years and zero otherwise. Rank and campus enter as sets of fixed effects capturing persistent compensation differences across faculty positions and UC campuses. The variable `Year` is included as a continuous linear trend controlling for long-run wage growth and other macroeconomic factors affecting compensation over the sample period.

It is important to note that our dataset is a repeated cross-section rather than a panel. Because the administrative salary records do not contain individual identifiers, we cannot track the same professor across years. Each annual file represents a snapshot of who was employed in that year. As a result, our regression estimates changes in the distribution of wages across groups rather than individual wage trajectories.

Wage data exhibit substantial heteroskedasticity across ranks; we therefore use HC1 heteroskedasticity-robust standard errors throughout the analysis.

To test whether wage dispersion changed after 2022, we employ two additional approaches. First, we conduct Levene’s tests for equality of variances within each rank, comparing the pre-2022 and post-2022 periods. Levene’s test is robust to non-normal distributions and is therefore appropriate for wage data.

Second, we estimate a difference-in-differences specification to test whether wage gaps between ranks changed after 2022. This model includes interactions between the post-2022 indicator and faculty rank, allowing us to examine whether the post-period affected some ranks more strongly than others.

To verify that our model is not affected by severe multicollinearity, we compute Variance Inflation Factors (VIF) for the predictors. The VIF values for both the `Year` trend and the `post2022` indicator are approximately 2.2, well below commonly used thresholds of concern, confirming that collinearity does not distort the estimates.

Finally, we conduct a robustness check using $\log(\text{RegularPay})$ instead of $\log(\text{TotalWages})$ as the dependent variable. Because regular pay excludes bonuses and other forms of discretionary compensation, consistency across both measures indicates that the results are not driven by variation in supplemental pay components.

5 Results

5.1 Within-Rank Dispersion

Figure 2 in Appendix 7 shows the evolution of the coefficient of variation (CV) by faculty rank from 2012 through 2024. Appendix Table 5 reports the coefficient of variation by rank and year, from which the pre-2022 and post-2022 averages are summarized in the text.

Four of the five faculty ranks experienced a decline in relative wage dispersion over the sample period. The largest reduction occurs among Associate Professors (-4.61 pp), followed by Full Professors (-2.11 pp), Research Professors (-2.13 pp), and Clinical Professors (-0.86 pp). Assistant Professors are the only group that moves in the opposite direction: their CV increases by $+1.74$ pp from 2022 onward.

The CV patterns for Full and Associate Professors display a gradual decline that begins prior to 2022, suggesting that the compression within ranks developed progressively rather than appearing as a sharp break following the policy changes.

5.2 Baseline OLS Results

Appendix Table 6 reports the results of the baseline OLS regression with $\log(\text{TotalWages})$ as the dependent variable. The coefficient on the post-2022 indicator equals -0.0634 ($\text{SE} = 0.0063$, $p < 0.001$). Interpreted in percentage terms, this implies that wages from 2022 onward are approximately 6.3% lower than the level predicted by the long-run trend, conditional on rank and campus.

All rank coefficients are positive and statistically significant relative to the Assistant Professor baseline. The ordering of the coefficients confirms the expected compensation hierarchy within the UC system: Clinical $>$ Full $>$ Research $>$ Associate $>$ Assistant.

The estimated coefficient on the linear year trend is 0.0475, indicating approximately 4.75% annual nominal wage growth along the pre-period trend.

Replacing $\log(\text{TotalWages})$ with $\log(\text{RegularPay})$ produces a post-2022 coefficient of -0.0410 ($\text{SE} = 0.0043$, $p < 0.001$). The smaller magnitude suggests that part of the decline in total wages from 2022 onward reflects reductions in variable pay components such as bonuses or overload compensation. However, the direction and statistical significance of the result remain unchanged.

5.3 Variance and DiD Results

Levene’s tests for equality of variance (Appendix Table 9) indicate that absolute salary variance increased after 2022 across all five ranks. The variance ratios range from 1.36 to

1.61 and are statistically significant ($p < 0.001$).

At first glance this appears inconsistent with the CV compression results. The two findings are nonetheless compatible. While the absolute spread of wages increased, mean salaries grew faster than the spread for most ranks. As a result, relative dispersion measured by the coefficient of variation declined even though the dollar variance increased.

The difference-in-differences specification reported in Appendix Table 8 examines whether wage gaps between ranks changed after 2022. The baseline post-2022 coefficient equals -0.0813 ($SE = 0.0103$, $p < 0.001$) for Assistant Professors.

The interaction term for Full Professors is -0.0016 ($p = 0.888$), which is economically negligible and statistically indistinguishable from zero. This indicates no evidence that the wage gap between Full and Assistant Professors narrowed in the post-period.

Associate Professors exhibit a positive and statistically significant interaction ($+0.0476$, $p = 0.0002$), indicating that their wages declined less relative to trend than those of Assistant Professors. Clinical Professors also display a positive interaction ($+0.0548$, $p = 0.002$). The Research Professor interaction is positive but statistically insignificant ($+0.0301$, $p = 0.253$).

Within-rank regressions (Appendix Tables 11–15) show that the post-2022 wage shortfall relative to trend is largest for Assistant Professors (-9.1% , $p < 0.001$) and Full Professors (-7.0% , $p < 0.001$), smaller for Associate Professors (-4.4% , $p = 0.001$), and statistically insignificant for Clinical (-2.7% , $p = 0.247$) and Research ($+0.9\%$, $p = 0.805$) Professors.

6 Discussion

The results are broadly consistent with the theoretical predictions. Four of the five ranks show declines in the coefficient of variation from 2022 onward, with Associate Professors displaying the strongest compression signal (-4.61 pp). These patterns align with the prediction in Cullen & Pakzad-Hurson (2022) that more salient wage comparisons can lead employers to narrow compensation dispersion within peer groups.

At the same time, the difference-in-differences results indicate that the wage gap between ranks did not narrow in the post-period. The interaction term for Full Professors is effectively zero (-0.0016 , $p = 0.888$), suggesting that the compensation hierarchy between Full and Assistant Professors remained largely unchanged.

Taken together, these results imply that the compression pattern operated primarily within ranks rather than across them. Faculty are most likely to compare their compensation with that of colleagues occupying similar positions. An Assistant Professor does not evaluate their pay relative to a Full Professor in the same way that they would compare themselves to other Assistant Professors.

The Levene tests also highlight an important distinction between absolute and relative measures of inequality. Although absolute variance increased across all ranks after 2022, relative dispersion measured by the CV declined for most senior ranks because average salaries increased faster than the spread of the distribution.

Pre-2022 Compression Trajectory. The CV trends for Full and Associate Professors show gradual compression that begins well before 2022, which predates the post-period indicator. This pattern is a known limitation of a binary step-function design: the *post2022* dummy cannot distinguish between an acceleration of an existing trend and a level shift originating at that threshold. The paper’s contribution is therefore best understood as documenting a salary compression trajectory that was already underway and that intensified in the post-2022 period—rather than identifying the 2022 events as the origin of compression. Future work employing an event study or a continuous time-trend interaction would more cleanly characterize whether the rate of compression changed discontinuously around 2022.

Limitations. Several limitations should be noted. The data are a repeated cross-section rather than a true panel, meaning we cannot track individual faculty members over time. Consequently, we cannot distinguish between wage adjustments for existing faculty and changes in the composition of faculty within ranks. The regression R^2 values across specifications range from approximately 0.05 to 0.21. These modest values are expected given the degree of cross-sectional wage heterogeneity within ranks and campuses, and reflect the fact that rank and campus fixed effects with a linear time trend capture the broad structure of compensation without fully accounting for individual-level variation; they do not indicate model misspecification.

The decline in the mean salary observed in 2024 (-6.4%) coincides with a drop in the number of observations from 20,176 to 18,898. This raises the possibility that part of the observed change reflects compositional shifts rather than direct adjustments to wage policies. The observation drop could stem from a data reporting lag in the administrative source, faculty attrition or retirement, or a shift in the composition of ranks recorded in that year; the paper cannot distinguish among these explanations, and results for 2024 should be interpreted with corresponding caution.

Finally, the post-2022 indicator captures any structural change that occurred at that threshold. While the transparency legislation and labor actions provide a plausible explanation, other macroeconomic or institutional factors could also contribute to the observed patterns. For this reason the results should be interpreted as descriptive associations rather than causal estimates.

7 Conclusion

This paper examined within-rank salary dispersion among UC faculty from 2012 to 2024. Mean nominal wages grew 62.5 % before declining 6.4 % in 2024. The CV declined for Associate (−4.6 pp), Full (−2.1 pp), Research (−2.1 pp), and Clinical (−0.9 pp) professors from 2022 onward, while Assistant Professor CV rose +1.7 pp. The baseline OLS finds log wages approximately 6.3 % below trend post-2022. The DiD model finds no narrowing of the Full–Assistant wage gap, though Associate and Clinical professors fared relatively better.

These findings suggest that wage transparency and labor actions are associated with relative salary compression *within* established peer groups but have a more limited impact *across* structurally distinct positions. Internal equity pressures appear to operate primarily among direct peers who observe and compare their wages, while entry-level faculty compensation remains tethered to external market conditions. Future research with individual-level panel data would be needed to separate deliberate compression from compositional turnover.

References

- Bandiera, O., Barankay, I., & Rasul, I. (2005). Social preferences and the response to incentives: Evidence from personnel data. *The Quarterly Journal of Economics*, 120(3), 917–962.
- Fehr, E., & Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics*, 114(3), 817–868.
- Card, D., Mas, A., Moretti, E., & Saez, E. (2012). Inequality at work: The effect of peer salaries on job satisfaction. *American Economic Review*, 102(6), 2981–3003.
- Cullen, Z. B., & Pakzad-Hurson, B. (2022). Equilibrium effects of pay transparency. *Econometrica*, 91(3), 765–802.
- University of California. (2025). *UC Employee Headcount*. University of California Information Center. Retrieved from <https://www.universityofcalifornia.edu/infocenter>

Appendix A: Mathematical Model of Inequality Aversion

To formalize the theoretical mechanism, we draw on the inequality aversion framework from Fehr & Schmidt (1999). The agent’s utility is:

$$U_i = W_i - C(e_i) - \lambda \max(0, W_{\text{peer}} - W_i), \quad (3)$$

where W_i is the worker’s wage, $C(e_i)$ is the convex cost of effort, and the final term captures disutility from observing a higher-paid peer scaled by salience $\lambda \geq 0$.

Prior to 2022, UC salary data were publicly posted but received limited attention; we model this as $\lambda \approx 0$. The 2022 events shifted $\lambda > 0$ for a meaningful share of the workforce. Once $\lambda > 0$, the principal must either raise the lower-paid worker’s wage or reduce the higher-paid peer’s wage to maintain effort. Both channels narrow the within-group wage distribution, yielding the prediction of CV compression. The mechanism is limited to comparisons within a relevant reference group: an Assistant Professor does not compare their salary to a Full Professor’s because the rank difference is institutionally recognized as structurally justified.

Appendix B: Figures

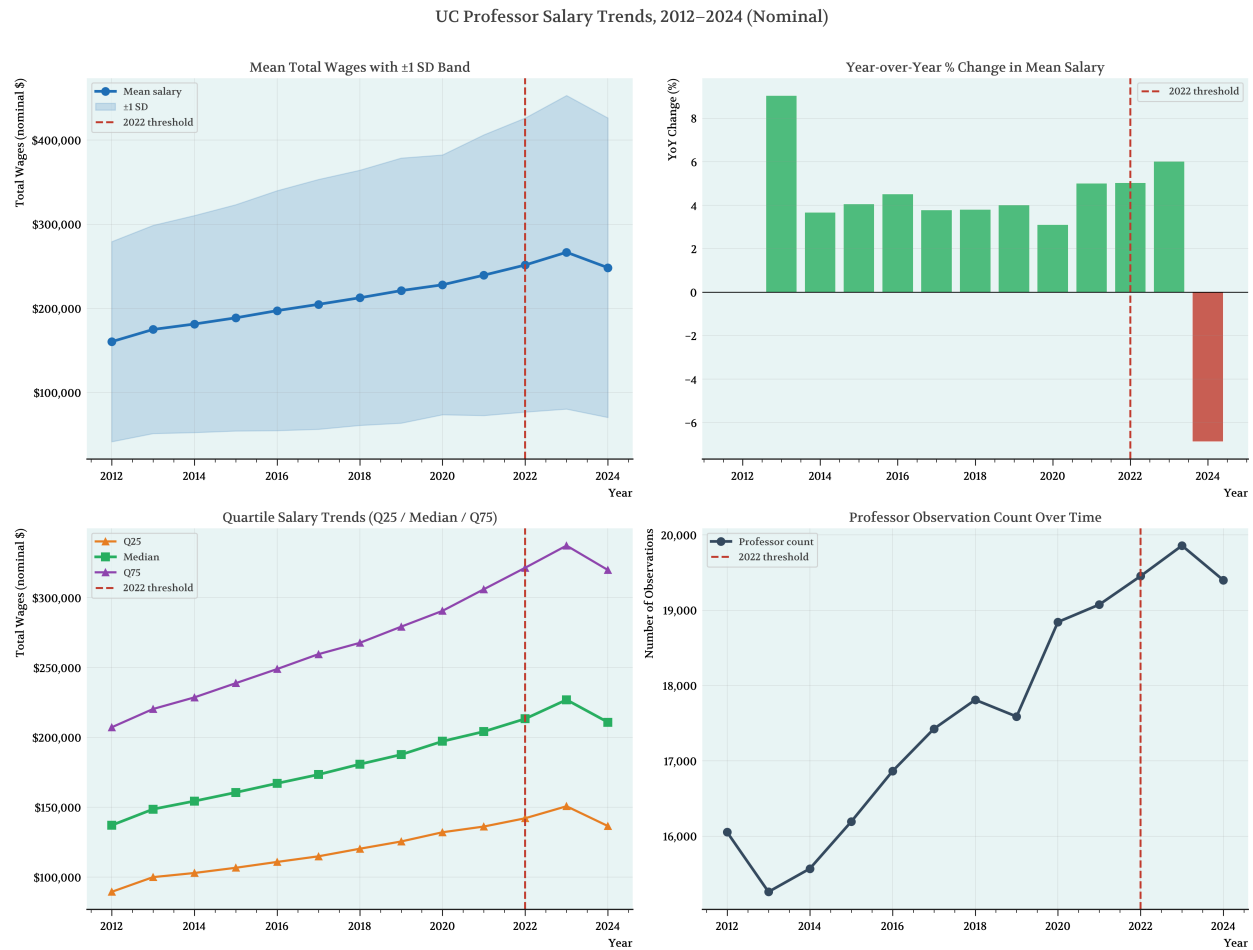


Figure 1: UC Professor Salary Trends, 2012–2024 (Nominal). Top left: mean total wages with ± 1 standard deviation band. Top right: year-over-year percentage change in mean salary; the 2024 bar (-6.4%) is the only negative value in the sample. Bottom left: quartile salary trends (Q25, median, Q75). Bottom right: total professor observation count by year. Vertical dashed line marks the 2022 threshold.

Salary Dispersion by Professor Rank, 2012–2024

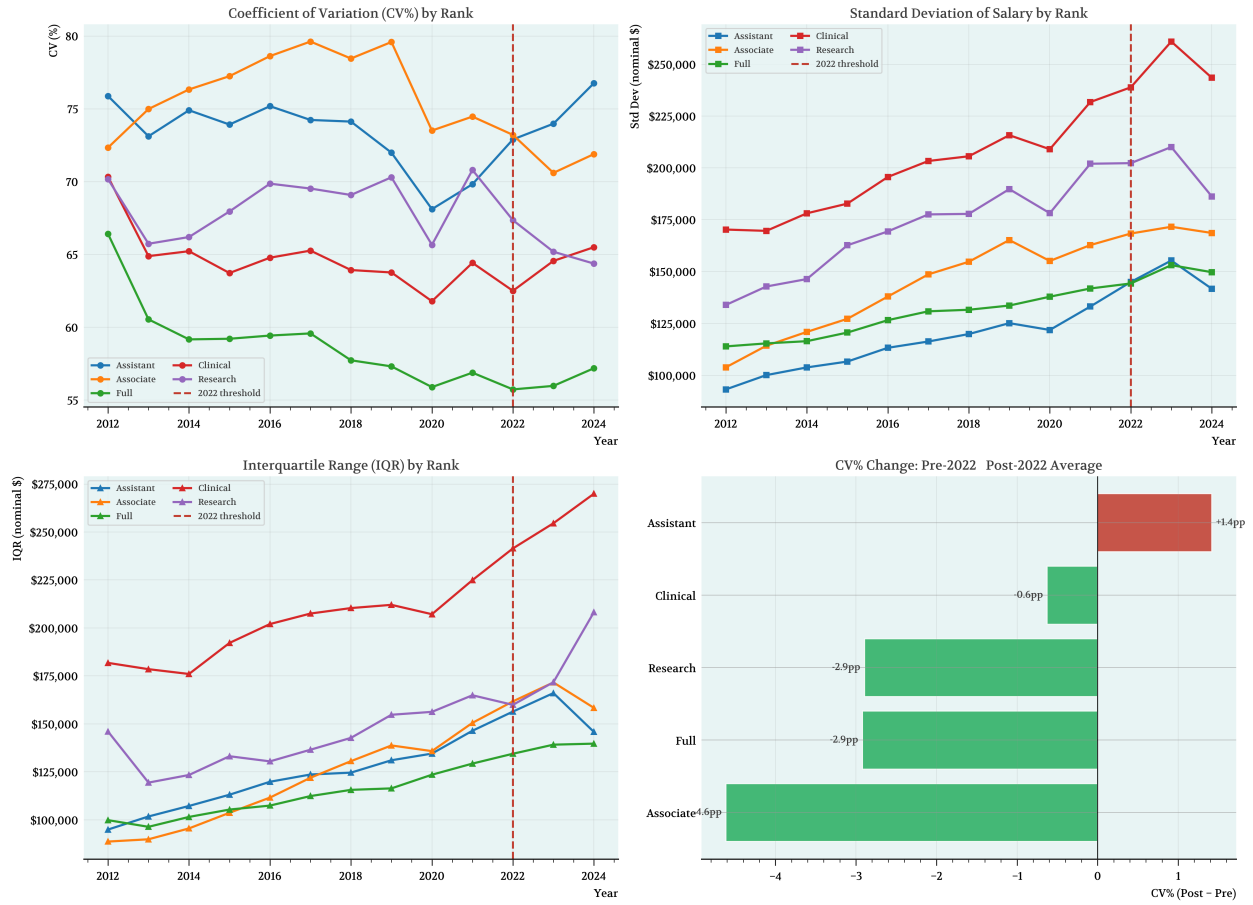


Figure 2: Salary Dispersion by Professor Rank, 2012–2024. Top left: coefficient of variation (CV%) by rank. Top right: standard deviation by rank. Bottom left: interquartile range by rank. Bottom right: bar chart of CV change in percentage points (post-2022 minus pre-2022 average). Vertical dashed line marks the 2022 threshold.

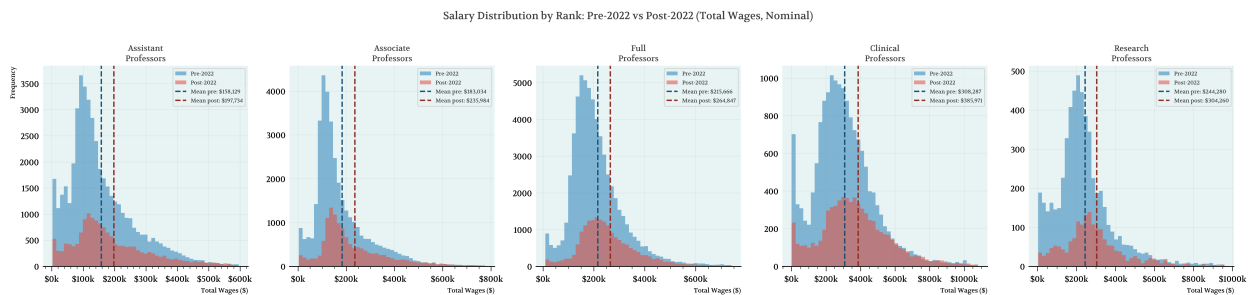


Figure 3: Salary Distribution by Rank: Pre-2022 vs. Post-2022 (Total Wages, Nominal). Each panel overlays the pooled pre-2022 (blue) and post-2022 (red) salary histograms for one rank. Vertical dashed lines mark the mean for each period.

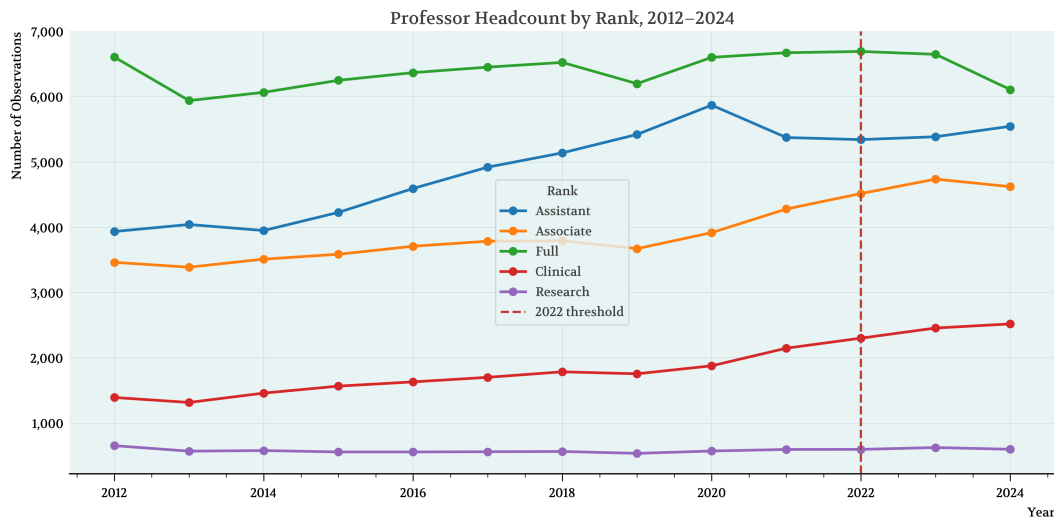


Figure 4: Professor Headcount by Rank, 2012–2024. Vertical dashed line marks the 2022 threshold.

Appendix C: Tables

Table: Coefficient of Variation (CV%) by Rank and Year

Year	Assistant	Associate	Full	Clinical	Research
2012	75.88	72.34	66.42	70.33	70.17
2013	73.12	74.99	60.54	64.89	65.74
2014	74.90	76.33	59.17	65.22	66.20
2015	73.92	77.25	59.21	63.73	67.95
2016	75.19	78.62	59.43	64.78	69.86
2017	74.24	79.62	59.58	65.27	69.53
2018	74.13	78.46	57.73	63.93	69.09
2019	71.99	79.59	57.31	63.76	70.30
2020	68.11	73.51	55.89	61.79	65.66
2021	69.83	74.47	56.88	64.42	70.81
2022	72.90	73.21	55.73	62.50	67.35
2023	73.98	70.60	55.97	64.56	65.19
2024	76.76	71.89	57.19	65.50	64.37

Coefficient of Variation (CV%) = $100 \times (\text{Std Dev} / \text{Mean})$. Computed from deduplicated panel.
Sample: professors with TotalWages > 0, years 2012–2024.

Table 5: Coefficient of Variation (CV%) by Rank and Year, 2012–2024. Computed as $100 \times \sigma/\mu$ within each rank-year cell.

Table 1. Baseline OLS: $\log(\text{Total Wages}) \sim \text{post2022} + \text{Rank FE} + \text{Campus FE} + \text{Year}$

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Associate***	0.2318	0.0057	40.983	0.0000	[0.2207, 0.2429]
Clinical***	0.6329	0.0084	75.106	0.0000	[0.6164, 0.6494]
Full***	0.5063	0.0049	103.550	0.0000	[0.4967, 0.5159]
Research***	0.4455	0.0119	37.518	0.0000	[0.4222, 0.4687]
post2022***	-0.0634	0.0063	-10.010	0.0000	[-0.0759, -0.0510]
Campus FE	Yes				
Year Trend	Yes				

N = 229,386 R² = 0.1024

Reference category: Assistant Professor.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Table 6: Baseline OLS regression of $\log(\text{Total Wages})$ on post2022, rank fixed effects, campus fixed effects, and a linear year trend. $N = 229,386$. HC1 robust standard errors. Reference category: Assistant Professor.

Table 2. Baseline OLS: $\log(\text{Regular Pay}) \sim \text{post2022} + \text{Rank FE} + \text{Campus FE} + \text{Year}$

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Associate***	0.2843	0.0038	75.125	0.0000	[0.2769, 0.2917]
Clinical***	0.6364	0.0052	122.608	0.0000	[0.6262, 0.6465]
Full***	0.6871	0.0035	197.242	0.0000	[0.6802, 0.6939]
Research***	0.5424	0.0091	59.335	0.0000	[0.5244, 0.5603]
post2022***	-0.0410	0.0043	-9.643	0.0000	[-0.0494, -0.0327]
Campus FE	Yes				
Year Trend	Yes				

N = 225,412 R² = 0.2164

Reference category: Assistant Professor.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Table 7: Robustness check: baseline OLS regression of $\log(\text{Regular Pay})$ on post2022, rank fixed effects, campus fixed effects, and a linear year trend. $N = 225,412$. HC1 robust standard errors. Reference category: Assistant Professor.

Table 3. DiD Compression: $\log(\text{Total Wages}) \sim \text{post2022} \times \text{C(Rank)} + \text{Campus FE} + \text{Year}$

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-84.3099	1.5467	-54.510	0.0000	[-87.3414, -81.2785]
post2022***	-0.0813	0.0103	-7.857	0.0000	[-0.1016, -0.0610]
Year***	0.0475	0.0008	61.954	0.0000	[0.0460, 0.0490]
Rank FE	Yes				
Campus FE	Yes				
post2022:Rank: Associate***	0.0476	0.0129	3.681	0.0002	[0.0223, 0.0730]
post2022:Rank: Clinical**	0.0548	0.0179	3.058	0.0022	[0.0197, 0.0899]
post2022:Rank: Full	-0.0016	0.0111	-0.141	0.8876	[-0.0233, 0.0202]
post2022:Rank: Research	0.0301	0.0263	1.143	0.2531	[-0.0215, 0.0817]

N = 229,386 $R^2 = 0.1026$

DiD spec: $\log_total_wages \sim \text{post2022} * \text{C(Rank)} + \text{C(EmployerName)} + \text{Year}$.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Reference category: Assistant professors.

Table 8: Difference-in-differences compression test. Dependent variable: $\log(\text{Total Wages})$. Specification includes post2022, $\text{post2022} \times \text{Rank}$ interactions, campus fixed effects, and a linear year trend. $N = 229,386$. HC1 robust standard errors. Reference category: Assistant Professor.

Table: Levene's Test for Variance Equality — Pre-2022 vs Post-2022

Rank	Pre-2022 Variance	Post-2022 Variance	Variance Ratio	Levene Stat	p-value	Significant
Assistant	1.372e+10	2.185e+10	1.5928	591.7232	4.1563e-130	Yes
Associate	2.069e+10	2.878e+10	1.3913	353.2237	1.5528e-78	Yes
Full	1.682e+10	2.226e+10	1.3238	288.7538	1.1953e-64	Yes
Clinical	4.117e+10	6.172e+10	1.4990	194.4820	4.9729e-44	Yes
Research	2.926e+10	4.012e+10	1.3713	36.3683	1.7100e-09	Yes

Levene's test for equality of variances (pre-2022 vs post-2022) within each professor rank.
Variance Ratio > 1: variance increased post-2022. Significant at $\alpha = 0.05$.

Table 9: Levene’s Test for Variance Equality—Pre-2022 vs. Post-2022. Variance Ratio > 1 indicates absolute variance *increased* post-2022. All five ranks show significant changes ($p < 0.001$), consistent with CV compression because mean salaries rose faster than the absolute spread for senior ranks.

Table: Variance Inflation Factors (VIF) — Baseline Regression Predictors

Predictor	VIF
Year	2.197
post2022	2.193
Rank	1.005
EmployerName	1.004

VIF computed using label-encoded Rank and EmployerName (approximation).
Threshold: VIF > 5 = moderate concern; VIF > 10 = severe multicollinearity.
Model: $\log_total_wages \sim post2022 + C(Rank) + C(EmployerName) + Year$ | HC1 robust SE.

Table 10: Variance Inflation Factors (VIF) for the baseline regression predictors. All VIFs are below the threshold of 5, confirming that multicollinearity between the Year trend and the post2022 indicator is not severe.

Within-Rank Regression: Assistant Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-88.6824	3.2485	-27.299	0.0000	[-95.0493, -82.3154]
Campus FE	Yes				
post2022***	-0.0954	0.0132	-7.226	0.0000	[-0.1213, -0.0696]
Year***	0.0496	0.0016	30.779	0.0000	[0.0464, 0.0527]

N = 63,756 R² = 0.0531

Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Assistant professors only.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Table 11: Within-Rank Regression: Assistant Professors. Specification: $\log(\text{TotalWages}) \sim \text{post2022} + C(\text{EmployerName}) + \text{Year}$. $N = 63,756$. HC1 robust standard errors.

Within-Rank Regression: Associate Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-86.7557	3.1266	-27.747	0.0000	[-92.8838, -80.6276]
Campus FE	Yes				
post2022***	-0.0452	0.0130	-3.468	0.0005	[-0.0708, -0.0197]
Year***	0.0488	0.0016	31.476	0.0000	[0.0458, 0.0518]

N = 50,988 R² = 0.0645

Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Associate professors only.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Table 12: Within-Rank Regression: Associate Professors. $N = 50,988$. Specification and standard errors as in Table 11.

Within-Rank Regression: Full Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-80.6041	2.1388	-37.687	0.0000	[-84.7960, -76.4121]
Campus FE	Yes				
post2022***	-0.0728	0.0086	-8.493	0.0000	[-0.0896, -0.0560]
Year***	0.0460	0.0011	43.361	0.0000	[0.0439, 0.0481]

N = 83,141 R² = 0.0523

Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Full professors only.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Table 13: Within-Rank Regression: Full Professors. $N = 83,141$. Specification and standard errors as in Table 11.

Within-Rank Regression: Clinical Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-85.1453	6.1922	-13.750	0.0000	[-97.2819, -73.0088]
Campus FE	Yes				
post2022	-0.0272	0.0235	-1.157	0.2473	[-0.0732, 0.0189]
Year***	0.0477	0.0031	15.530	0.0000	[0.0417, 0.0537]

N = 23,918 R² = 0.0785

Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Clinical professors only.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Table 14: Within-Rank Regression: Clinical Professors. $N = 23,918$. The post2022 coefficient is negative but statistically insignificant ($p = 0.247$), reflecting the distinct dynamics of health-system compensation.

Within-Rank Regression: Research Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-53.6338	8.7700	-6.116	0.0000	[-70.8226, -36.4449]
Campus FE	Yes				
post2022	0.0092	0.0375	0.246	0.8054	[-0.0642, 0.0827]
Year***	0.0324	0.0043	7.452	0.0000	[0.0239, 0.0409]

N = 7,583 R² = 0.0724

Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Research professors only.

HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Table 15: Within-Rank Regression: Research Professors. $N = 7,583$. The post2022 coefficient is positive and statistically insignificant ($p = 0.805$).